SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain classification of materials into conducting, semiconducting and insulating materials based on their energy band.

Q.24 Describe the thermocouple with diagram and mention its important applications.

Q.25 Explain various engineering materials for fabrication of transformer.

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3rd Sem / Electrical , Power station Engg., Elect. & Eltx. Engg., Fire Tech & Safety

Subject : Electrical Engineering Materials / Electrical and Electronics Engineering Materials

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Forbidden band is largest in

a) Conductor b) semiconductor

c) insulator d) super conductor

Q.2 Commonly used conducting materials are

a) Copper and aluminum

b) silver and gold

c) brass and bronze

d) carbon and nichrome

Q.3 _____ is used for winding of coils for dc motor starters.

a) constantan b) nichrome

c) copper d) aluminum

Q.4 High frequency transformer core are generally made from

a) Alnico b) ferrites

c) Mu-metal d) Silicon steel

Q.5 Soft magnetic material is

a) tungsten steel b) bismuth

c) iron d) Alcomax

Q.6 Varnish used in coating, impregnation and adhesion should have the property (ies) of

a) quick drying

b) chemical stability

c) Setting hard and with good surface

d) All of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The electrons present in the outermost orbit of an atom are known as free electrons. (True/False)

Q.8 Brass and bronze are alloys of______.

Q.9 Expand PVC

Q.10 Transformer bushing are made of_______

Q.11 CRGOS sheet is used for construction of cores of rotating electrical machines. (True/False)

Q.12 A fuse is a thin wire of material having high melting point. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 List the desirable properties of good conductor.

Q.14 Compare low resistivity and high resistivity materials.

Q.15 State the important properties and application of Tungsten.

Q.16 Differentiate between intrinsic and extrinsic semiconductor.

Q.17 Explain the electrical properties of insulating materials.

Q.18 Enlist the properties of insulating oil and epoxy resin.

Q.19 Give the properties and applications of mica.

Q.20 Classify the ferromagnetic material.

Q.21 Explain the properties of cobalt steel.

Q.22 State the applications of hard ferrites.